

assignment _momentum_impulse

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:42pm

Quiz Instructions

PLEASE. USE a post-it to hide the hints.

Take your time. Work with a group.



Question 1 8 pts

A car of mass 1689 kg collides head-on with a parked truck of mass 2000 kg. Spring mounted bumpers ensure that the collision is essentially elastic. If the velocity of the truck is 17 km/h (in the same direction as the car's initial velocity) after the collision, what was the initial speed of the car?

hint: you don't need to convert speeds since there is km/h on both sides of the equality.



38 km/h



29 km/h



19 km/h



10 km/h



Question 2 8 pts

A 5.00-kg ball is hanging from a long but very light flexible wire when it is struck by a 1.50-kg stone traveling horizontally to the right at 12.0 m/s. The stone rebounds to the left with a speed of 8.50 m/s, and the ball swings to a maximum height h above its original level. The value of h is closest to

hint: first use conservation of momentum. The velocity of the stone after the collision is negative. Find velocity of the ball after (positive. Watch the sign). Second, use conservation of energy. KE of the ball = potential energy of the ball (cancel the mass),



1.93 m.



3.69 m.



2.20 m.



0.0563 m.



1.10 m.



Question 3 8 pts

An 8.0-g bullet is shot into a 4.0-kg block, at rest on a frictionless horizontal surface (see the figure). The bullet remains lodged in the block. The block moves into an ideal massless spring and compresses it by 8.7 cm. The spring constant of the spring is 2400 N/m. The initial velocity of the bullet is closest to

hint: convert g to kg. Step1: use conservation of momentum (hit and stick situation). Find speed of block + bullet. Step2: Use conservation of energy. KE (of both bullet and block) = elastic potential energy. Conver cm to m.



1100 m/s.



1200 m/s.



1300 m/s.



900 m/s.



1000 m/s.



Question 4 8 pts

A 2200-kg auto moving northward at 12.0 m/s runs into a 3800-kg truck which is also moving northward, but at 5.00 m/s. If the vehicles lock bumpers, how fast are they moving just after the collision?

hint: Its a hit and stick situation but the velocities have the same sign before collision. After the move with

the same velocity $(m_1+m_2)V' = \text{momentum after}$

1.

☐

3.57m/s

☐

- 5m/s

☐

7.57 m/s

☐

12m/s



Question 5 8 pts

Two objects of the same mass move along the same line in opposite directions. The first mass is moving with speed v . The objects collide, stick together, and move with speed $0.100v$ in the direction of the velocity of the first mass before the collision. What was the speed of the second mass before the collision?

hint: They have the same mass (cross out). One mass has a negative velocity after. They stick to each other,

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1.20v

☐

0.800v

☐

10.0v

☐

0.00v

☐

0.900v



Question 6 8 pts

A 620-g object traveling at 2.1 m/s collides head-on with a 320-g object traveling in the opposite direction at 3.8 m/s. If the collision is perfectly elastic, what is the change in the kinetic energy of the 620-g object?

hint: Before the velocity is negative for one of the mass. you don't need to convert g to kg . After the collision the relative velocity is the same'(with negative velocity) .

So $V_1 - V_2 = V_2' - V_1' = 2.1 - (-3.8) = 5.9 = V_2' - V_1'$ (conservation of KE_

and $m_1V_1 + m_2V_2 = m_1V_1' + m_2V_2'$ (conservation of momentum)

Find V_1' and V_2' then change = $0.5 m_1 V_1'^2 - 0.5 m_1 V_1^2$

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It loses 0.23 J.

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It doesn't lose any kinetic energy because the collision is elastic.

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It loses 0.47 J.

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It gains 0.69 J.

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It loses 1.4 J.

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Question 7 8 pts

A 2.3-kg object traveling at 6.1 m/s collides head-on with a 3.5-kg object traveling in the opposite direction at 4.8 m/s. If the collision is perfectly elastic, what is the final speed of the 2.3-kg object?

hint: use conservation of momentum and relative velocity : $V_2 - V_1 = - (V_2' - V_1')$

If direction is opposite , the sign is opposite (-4.8). and $m_1V_1 + m_2V_2 = m_1V_1' + m_2V_2'$

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0.48 m/s

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3.8 m/s

☐

7.1 m/s

☐

4.3 m/s

☐

6.6 m/s

☐

Question 8 8 pts

During a collision with a wall, the velocity of a 0.200-kg ball changes from 20.0 m/s toward the wall to 12.0 m/s away from the wall. If the time the ball was in contact with the wall was 60.0 ms, what was the magnitude of the average force applied to the ball?

hint: remember change in momentum = $F \times \text{time}$. $V_i = 20$ and $V_f = -12$. convert ms to s

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13.3 N

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40.0 N

☐

107 N

☐

26.7 N



16.7 N



Question 9 8 pts

A golf club exerts an average horizontal force of 1000 N on a 0.045-kg golf ball that is initially at rest on the tee. The club is in contact with the ball for 1.8 ms. What is the speed of the golf ball just as it leaves the tee?

hint: The force is the same for both the ball and the club. Impulse = change in momentum. $V_i=0$
find V_f



40 m/s



45 m/s



35 m/s



30 m/s



50 m/s



Question 10 8 pts

Astronaut Jennifer's lifeline to her spaceship comes loose and she finds herself stranded, "floating" 100 m from the mothership. She suddenly throws her 2.00-kg wrench at 20 m/s in a direction away from the ship. If she and her spacesuit have a combined mass of 200 kg, how long does it take her to coast back to her spaceship?

hint: Its a recoil situation. Find the speed of the astronaut. Then use kinematics to find how long it takes to cover the 100m (there is no friction in space)



1000 s



250 s



500 s



750 s



2.50 min



Question 11 8 pts

Two ice skaters suddenly push off against one another starting from a stationary position. The 45-kg skater acquires a speed of 0.375 m/s relative to the ice. What speed does the 60-kg skater acquire relative to the ice?

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0.75 m/s

☐

0.50 m/s

☐

0.00 m/s

☐

0.28 m/s

☐

0.38 m/s



Question 12 8 pts

A 1200-kg cannon suddenly fires a 100kg cannonball at 35m/s What is the recoil speed of the cannon? Assume that frictional forces are negligible and the cannon is fired horizontally.

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35 m/s

☐

2.9 m/s

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3.2 m/s

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3.5 m/s



Question 13 8 pts

A 0.14-kg baseball is dropped from rest from a height of 2.0 m above the ground. What is the magnitude of its momentum just before it hits the ground if we neglect air resistance?

hint: find the speed before it hits the floor (use conservation energy. $PE=KE$). Multiply mass x final velocity.

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0.62 kg · m/s

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0.28 kg · m/s

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0.88 kg · m/s

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0.44 kg · m/s

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1.4 kg · m/s

Not saved

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